

Setting Up a Test for a Population Proportion

AP Statistics · Topic 6.4 · Teacher Edition

OBJECTIVES

Math Objective	Set up a one-proportion z -test by (1) stating H_0 and H_a using correct parameter notation p , (2) verifying all three conditions (Random, 10%, Large Counts), and (3) computing the test statistic $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$.
Language Objective	Students will use the frame “If the true proportion is _____, the probability of getting a sample proportion as extreme as _____ by chance alone is _____.” to interpret a p -value in context.
Essential Understanding	A significance test answers ONE question: is the observed sample proportion far enough from p_0 that we can reject the null? The three-condition check is not busywork — it validates that the z -statistic formula is trustworthy for this sample.

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Students move from informal “is this surprising?” thinking (Do Now) to the formal four-step AP test structure. Explore runs the full hypothesis-testing sequence on one continuous context so each step builds on the last.
Essential Question	How do we decide whether an observed sample proportion is statistically surprising, and what conditions must hold before we can trust that decision?
Materials	Pencils; student packet; calculator (normalcdf / z -table for optional reinforcement). No Chromebooks required during Explore.

[PIN] PRE-FLIGHT — SINGLE-PERIOD QUICK: DOK-3 SPINE INSIDE ONE PERIOD

Today’s conceptual core is Practice #4 (DOK-3 Performance Task). Students design their own scenario, state hypotheses, and justify conditions. That design step IS the deepest content.

Pace: Do Now (7 min) → Launch (10 min) → Explore #1–3 (18 min) → DOK-3 #4 (10 min) → Share (5 min) → Exit (5 min). ~55 min total.

If running tight: cut Explore #2 (conditions check can be done as teacher-led rapid fire). Never cut #4.

Tag: bridge from sampling distributions

Do Now

DOK: 1

Minutes: 7

Teacher does

- Hand out at door.
- Circulate 3 min.
- Cold-call 1 student for $\hat{p}$ computation.
- Bridge aloud: ‘Today we make that informal intuition formal.’

Students do

- Compute $\hat{p}$ from given counts.
- Compare to stated p_0 and write one-sentence informal judgment.

Questions to ask

- What does $\hat{p}$ represent vs. p ?
- How far from p_0 is 'far enough' to be suspicious?

Adult role

Solo teacher: circulate, time 7 min, pull sheets.

Do Now

A coin is flipped 80 times and lands heads 52 times. (a) What is $\hat{p}$, the sample proportion of heads? (b) If the coin were fair, what would you expect $\hat{p}$ to equal? (c) Does your result from (a) convince you the coin is unfair? Explain in one sentence using symbolic form.

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

Tag: model the four-step test structure

Launch**DOK: 2****Minutes: 10****Teacher does**

- Write the four AP steps on board: Hypotheses / Conditions / Test Statistic / Conclusion.
- Walk through the SETUP RULES callout completely.
- Model only Step 1 (hypotheses) on one fresh example out loud.

Students do

- Copy the four-step framework into margin.
- Fill in the hypothesis example as teacher models.

Questions to ask

- What is the null hypothesis ALWAYS claiming about p ?
- When do we use $<$, $>$, or $\neq$ in H_a ?

Adult role

Teacher models ONE step only. Students finish the rest in Explore.

[BOOK] SETUP RULES: ONE-PROPORTION z -TEST (keep printed on your page)

Four AP Steps (write these before every significance test):

1. **Hypotheses:** State $H_0: p = p_0$ and $H_a: p \neq p_0$ (or $<$, $>$). Always about the *parameter* p .
2. **Conditions:** (i) Random sample/assignment. (ii) Independence: $n < 10\%$ of population. (iii)

Large Counts: $np_0 \geq 10$ and $n(1 - p_0) \geq 10$.

3. **Test Statistic:** $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$. Use p_0 , NOT $\hat{p}$, in the denominator.

4. **Conclusion:** State whether you reject or fail to reject H_0 , in context, at the given α level.

[WARN] COMMON ERRORS (both appear on AP Free Response!)

ERROR 1: Writing $H_0: \hat{p} = p_0$. The null is about the *parameter* p , never the *statistic* $\hat{p}$.

ERROR 2: Using $\hat{p}$ instead of p_0 in the standard error denominator. The formula assumes the null is true.

ERROR 3: Concluding “we proved H_0 ” when you fail to reject. We only ever *fail to reject* — never accept H_0 .

Tag: TPS + DOK-3 driver

Explore

DOK: 2-3

Minutes: 28

Teacher does

- 4 laps circulation (~7 min each).
- Lap 1: #1 (hypotheses). Lap 2: #2 (conditions). Lap 3: #3 (z-stat). Lap 4: #4 (DOK-3).
- Do NOT give DOK-3 answers. Pairs present at Share.

Students do

- Work with partner, in order.
- For #4, choose their own real-world claim before writing a single symbol.

Questions to ask

- #2: which condition is hardest to verify in practice? Why?
- #4: why does your scenario call for one-sided vs. two-sided?

Adult role

Press pairs on #4 direction choice. That IS the DOK-3 judgment.

Practice #1 — State the hypotheses

A city council claims that 60% of residents support a new park. A survey of 120 randomly selected residents finds 64 who support the park. State the null and alternative hypotheses (using parameter p) for a two-sided test of the council's claim.

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

Practice #2 — Check all three conditions

Using the park-survey context from Practice #1 ($n = 120$, $p_0 = 0.60$): verify all three conditions for a one-proportion z -test. Show each check symbolically.

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

Practice #3 — Compute the test statistic z

Still using the park-survey context ($\hat{p} = 64/120$, $p_0 = 0.60$, $n = 120$): compute the test statistic using the formula $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$. Leave intermediate values in symbolic form until the final step.

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

Practice #4 [STAR] PERFORMANCE TASK — Design your own 1-prop test

[STAR] PERFORMANCE TASK: Choose a real-world claim about a population proportion (e.g., school attendance rate, recycling rate, seat-belt use). (A) State H_0 and H_a with the correct parameter p and direction. (B) Describe a realistic sample design and check whether all three conditions would plausibly be met. (C) Explain why a researcher would choose a one-sided vs. two-sided alternative for your scenario.

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

Tag: DOK-3 pair presents + bridge to p-value

Share Summary**DOK: 2****Minutes: 5****Teacher does**

- Cold-call a pair that struggled on #4 direction choice.
- Bridge: 'Tomorrow we compute the p -value and make a formal conclusion.'

Students do

- Presenting pair uses the sentence frame for #4 direction justification.
- Rest of class signals thumbs up/sideways and writes takeaway.

Questions to ask

- What would make you change from two-sided to one-sided?
- Preview: if $p\text{-value} < \alpha$, what do we say about H_0 ?

Adult role

Cold-call the pair that had the most interesting scenario — hold them accountable.

[SEARCH] BRIDGE TO NEXT LESSON

Next class we compute the p -value (using a z -table or normalcdf) and write the full four-step conclusion. The setup you practiced today is the foundation. If you are unsure of any step, re-read the Setup Rules callout before next class.

Sentence frames

Direction justification: "I chose a (one/two)-sided alternative because I wanted to detect whether the proportion was specifically than p_0 , not just different."

Conditions check: "The Large Counts condition is met because $np_0 = \geq 10$ and $n(1-p_0) = \geq 10$."

Tag: summary recap (not graded)

Exit

DOK: 1-2

Minutes: 5

Teacher does

- Collect at door.
- Scan #3 (students who can't set up the formula). Plan tomorrow's Do Now if many are stuck.

Students do

- 3 sentence stems, honest.

Questions to ask

- Did students correctly identify the direction of H_a ?

Adult role

Collect. No grade.

The null hypothesis for a one-proportion z -test always states

The condition I found hardest to verify today was because

I distinguish a one-sided from a two-sided alternative by

OPTIONAL REINFORCEMENT (not collected)

Optional #1 — Interpret the p -value in context

The park-survey test yields $z \approx -1.49$ and p -value ≈ 0.136 . Interpret this p -value in context using the sentence frame: “If the true proportion of residents who support the park is ____, the probability of getting a sample proportion as far from ____ as ____ by chance alone is ____.” Then state your conclusion at the $\alpha = 0.05$ level.

[CHECK] ANSWER / ANTICIPATED TALK

(no answer key — verify before class)

[CLOCK] PERIOD VARIANTS — 55-MIN vs 65-MIN

55 min (standard): Run as written above.

65 min (extra period): Add 10 min to Explore — pairs share DOK-3 scenarios with a second pair (gallery-walk style) before whole-class share. Richer DOK-3 payoff.

45-min short day: Cut Explore #2 (run conditions check as teacher-led rapid fire, 2 min). Keep #4.

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Setup Rules callout as a laminated reference card.
- For #4, provide a pre-structured template: “Claim: ___ % of ___ do/are ___. $H_0: p = \underline{\hspace{1cm}}$. $H_a: p \neq \underline{\hspace{1cm}}$.”
- Accept verbal sentence-frame completion in lieu of written for exit ticket.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Word cards: “parameter” = true value for the whole population; “statistic” = computed from the sample.
- “Null hypothesis” = the boring claim (nothing unusual is happening).
- “Fail to reject” $\neq$ “accept” — display both phrases side-by-side.
- Pair ELL students with English-dominant partners for DOK-3 scenario brainstorm.